Classroom notes

The Day I Tried to Run on Pure Sparkles · for teachers, librarians and reading-group leaders

At a glance

Grades	1–3 (ages 6–9). Works as a read-aloud from Grade 1; strongest for independent reading at ages 7–9.
Topic	Sugar and blood sugar — and the habits of a scientist: observing, asking why, fair tests, data.
Story	Eight short chapters, about 1,300 words — roughly 15 minutes read aloud.
Back matter	Lumi's Secret Lab Notes (pp. 36–45) · Think Like a Scientist (pp. 46–47) · Lumi's Word Lab, 17 words (pp. 48–57) · Jiwoo's "Prove It" Tracker (pp. 58–63) · For the Grown-Up Lab Partner (pp. 64–66)

Big ideas

- Bodies respond to food — and responses differ from person to person.
- When, how much and with what can matter as much as what.
- Scientists observe before they decide. "No change" is a real result.
- A fair test changes one thing at a time.
- Labels and ads make promises; tests check them.

How this book handles food — please keep it this way in class

- No food is called good or bad, and there are no calorie or sugar-gram counts. Please keep it that way in class.
- No one's body is discussed. Keep comments on feelings and energy, never on size or weight.
- Don't collect, display or grade students' food records. The tracker belongs to the child and stays at home.
- Plan activities without food or drink, and don't use sweets as rewards during the unit.
- Praise "I didn't feel anything" and "I don't know." Let a wrong reason stand for a minute; ask one question instead of correcting it.

Sensitivity notes

Diabetes. Some students live with type 1 diabetes and take insulin. Follow their care plan and their family's lead, and never ask them to explain their condition to the class. If it comes up, a simple, accurate line: "Some people's bodies don't make enough insulin, so they take it as medicine. It's the same helper."

Food access and allergies. Not every student eats breakfast every day. Present "food before the drink" as one of Lumi's ideas to test, not a rule. Keep every activity food-free.

Body talk. If a student comments on someone's body, redirect to what bodies do: "In this book we're scientists of how bodies feel and work."

Three short lessons

Lesson 1 · Read & Notice *40 min*

5 min	Cover prediction: "What is this bottle promising?" Record predictions on the board.
15 min	Read aloud. Stop after Chapter 3 (predict), Chapter 5 (what changed?) and Chapter 8 (Jiwoo's three lessons).
10 min	Talk: choose two or three discussion questions.
7 min	Introduce Think Like a Scientist (p. 46): Notice · Wonder · Test · Report.
3 min	Exit ticket: one thing I noticed, one thing I wonder.

Lesson 2 · Model It *40 min*

10 min	Re-read Lab Notes 1–3 (pp. 39–41): the roller coaster, the Insulin Crew, two grabbers.
20 min	Activity: the Insulin Crew Relay (active) or Draw the Roller Coaster (seated).
10 min	Debrief: explain "two grabbers" in your own words. Vocabulary: glucose, insulin, pancreas, energy.

Lesson 3 · Test It *40 min*

15 min	Activity: Four Things Were True at Once (card sort).
15 min	Data detectives: analyse Jiwoo's filled-in week (pp. 61–62) with the questions below.
10 min	Write a fair test on paper. Optional: send home the blank tracker with the family letter.

Data detectives — Jiwoo's week (pages 61–62)

Jiwoo's week is made-up data, which makes it safe to analyse together — no student shares their own food. Ask the questions in order; teacher notes on the right.

Question	Teacher note
On how many days did Jiwoo have something sweet?	Six — Day 6 is blank. She only noticed because she wrote it down (p. 62).
On which days did she feel nothing?	Days 2, 3 and 5. Three of six logged days show nothing — on purpose (p. 65).
What was different about Day 1?	A lot of sugar, all at once, on an empty stomach, right before running — the four things on p. 42.
On Days 2 and 3 she had something sweet with or after food and felt nothing. Does that prove food stops a crash?	Not yet — one kid, a few days. It's a pattern worth testing, one change at a time.
Day 4 is an open case: was the homework fog a sugar crash, or the Ice Age? Why is writing down both smart?	A scientist records every possible explanation, then tests one. (The Ice Age is a joke — ask students how they'd rule it out.)
What would Jiwoo need to do to be more sure?	More days, and change one thing at a time while keeping the rest the same.

Science background for teachers

Blood sugar has a range. Glucose from carbohydrates enters the blood after eating. The pancreas releases insulin, which signals muscle, fat and liver cells to take glucose in; when blood sugar falls, a second hormone, glucagon, helps release stored glucose. The Insulin Crew is a model of the first half — insulin is a signal, not a carrier, as Lumi admits on page 40.

Two grabbers. Working muscles take glucose from the blood by a route that doesn't need insulin. Insulin and exercise together clear glucose faster than either alone — the idea behind Lumi's "two grabbers at once."

The dip happens to some, not all. In studies of trained adult athletes, a large sugary drink 30–60 minutes before exercise lowered blood sugar early in exercise in some people but not others, and feeling weak rarely matched a measured low. There are no comparable studies in healthy children. Jiwoo's day is one case — which is why the book asks children to observe rather than predict.

Food first. Fat, protein and fibre slow how fast the stomach empties, so the same carbohydrate eaten with or after food raises blood sugar more gently (studies in adults).

When it's more than a dip. Low blood sugar that needs treatment is rare in children without diabetes. Shakiness, sweating, dizziness or confusion that keep coming back are a reason to call the pediatrician — as is much more thirst and urination with tiredness or weight loss, a warning sign of type 1 diabetes.

Sugar and behaviour. Controlled studies have not found that sugar changes children's behaviour or thinking, but adults who expect a sugar effect tend to see one — one more reason to ask children what they noticed instead of telling them.

Standards connections

Short descriptions in our own words — check the official documents for exact wording. An honest note first: in U.S. science standards, the body's inner organs first appear in Grade 4 and energy from food in Grade 5. At Grades 1–3, this book supports the practices of science rather than a body or nutrition standard.

Science practices (Grades K–2 and 3–5)

Practice	What it looks like	Where it shows up
Asking questions	Ask questions you can test — in Grades 3–5, ask what would happen if one thing changed.	Four Things Were True at Once
Using models	Use a model to show how something works — and notice how the model differs from the real thing.	Insulin Crew Relay · Draw the Roller Coaster
Fair tests	Plan a fair test that changes one thing at a time.	Four Things · My Case Report
Working with data	Record observations, look for patterns and compare them with predictions.	Data detectives · trackers
Explaining with evidence	Build an explanation from what you observed.	Discussion · Label Detective
Cause and effect	Things that happen together are not always cause and effect.	Four Things Were True at Once
Systems	The parts of a system work together.	Insulin Crew Relay

Common Core State Standards — English Language Arts

Code	In short	Where it shows up
RL.1.3 · RL.2.3 · RL.3.3	Describe characters, how they respond to events, and how their actions move the story.	Read-aloud and discussion
RL.2.7 · RL.3.7	Use the pictures to understand characters, setting and plot — and what the art adds to the words.	Read-aloud picture walk
RI.2.3 · RI.3.3	Describe how scientific ideas connect, using sequence and cause-and-effect language.	Lab Notes · Roller Coaster · Relay
RI.2.4 · RI.3.4 · L.2.4 · L.3.4	Work out the meaning of subject words and phrases.	Word Lab Workout
SL.1.1 · SL.2.1 · SL.3.1	Take part in conversations about texts with partners and groups.	Discussion · Four Things · Label Detective

Codes refer to the Common Core State Standards for English Language Arts. © Copyright 2010. National Governors Association Center for Best Practices and Council of Chief State School Officers. All rights reserved.

Health education skills

Skill	What it looks like	Where it shows up
Influences	Notice what influences health choices — including media and labels.	Label Detective
Checking claims	Ask whether health information and product claims can be trusted.	Label Detective
Asking for help	Say how you feel, and know when to tell a trusted adult.	Discussion · trackers
Decisions	Compare options, then reflect on how a choice turned out.	Discussion · Data detectives

IB Primary Years Programme

Theme	Why it fits	Where it shows up
Who we are	An inquiry into the nature of the self, including physical and mental health.	The whole unit

Vocabulary — Lumi's Word Lab

Words for every case: case study · subject · data · observe · hypothesis · experiment

Words from this case: energy · fuel · carbohydrate · glucose · blood sugar · empty stomach · sugar spike · pancreas · insulin · cell · sugar crash

When a student says...

"So soda is bad for you."	"What was different about that day?" (the book's own question, p. 64)
"I felt nothing."	"Good — you wrote down what actually happened. That's the whole job."
"Can we test it in class?"	Not with food. We test on paper here; at home, it's the family's choice, one week only.
"My cousin has diabetes."	"Thanks for sharing. Bodies work differently, and doctors help people take care of theirs." Move on gently.

Family letter

Copy onto school letterhead, or print the one-page version from the hub.

Dear families,

This week our class read The Day I Tried to Run on Pure Sparkles, a story about a soccer player who learns to observe her own body like a scientist.

Your child may bring home a one-week Sweet Stuff Experiment. It is optional and private: nothing is collected, checked or graded. It asks your child to notice when they have something sweet, and how their body feels 30 minutes later — and it treats "I felt nothing" as a real result.

If you'd like to help: ask, don't check. Praise "I didn't feel anything." Don't say the finding first. And stop after one week.

The book never calls a food good or bad and includes no calorie or sugar counts; we're keeping it that way. If you have questions about your child's health, please talk with your pediatrician.

More for families: lumisnutritionlab.com

Corrections to Lumi's Word Lab

Three entries in Lumi's Word Lab simplify too far. Here is what each one says in the book, and what it should say. The rest of the Word Lab stands.

CARBOHYDRATE (page 53) — in the book: Some carbs break down slowly (like rice), others break down fast (like juice).

> It should say: Some carbs raise your blood sugar slowly (like beans or apples). Others raise it fast (like white bread or white rice).
>
> In the international glycemic index tables, white rice is a fast carb, and orange juice and apple juice are slower ones — the book's example was backwards. Sources 23, 24.

GLUCOSE (page 54) — in the book: Almost everything you eat eventually becomes glucose before your body burns it.

> It should say: Your body turns most carbs into glucose, a sugar it burns for energy. It can burn fat for energy too, and it uses protein mostly to build and fix your body.
>
> Fat is burned directly — only a small part of it can become glucose — and protein is used mostly for building. Source 2.

CELL (page 57) — in the book: …trillions of them, all needing glucose to survive.

> It should say: …trillions of them. Every cell needs fuel. Many can burn fat too, but some, like red blood cells, need glucose.
>
> "Trillions" is right — about 17 trillion in a 10-year-old. But heart and muscle cells, among others, also burn fat; red blood cells depend on glucose. Sources 2, 25, 26.

References

Every source behind the science in this book — what each one supports, and who was studied. Most of the research is in adults; we say so under each one. Nothing here is medical advice.

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   A free textbook. Supports how insulin and glucagon work, why fat and protein slow the stomach, how the body burns fat and protein, and which cells need glucose (glossary).
3. What is diabetes? National Institute of Diabetes and Digestive and Kidney Diseases. Reviewed April 2023. Accessed November 4, 2026. www.niddk.nih.gov/health-information/diabetes/overview/what-is-diabetes
   Supports the idea that insulin helps glucose get into your cells so they can use it for energy.
4. Richter EA, Hargreaves M. Exercise, GLUT4, and skeletal muscle glucose uptake. Physiol Rev. 2013;93(3):993-1017. doi:10.1152/physrev.00038.2012. doi.org/10.1152/physrev.00038.2012
   Supports how working muscles open sugar "doors" (GLUT4) when they contract. Review article.
5. Sylow L, Kleinert M, Richter EA, Jensen TE. Exercise-stimulated glucose uptake: regulation and implications for glycaemic control. Nat Rev Endocrinol. 2017;13(3):133-148. doi:10.1038/nrendo.2016.162. doi.org/10.1038/nrendo.2016.162
   Supports the point that exercising muscles take in sugar by a different route from insulin. Review article.
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   Supports the finding that insulin and exercise together clear more sugar from the blood than either one alone. Study in 10 healthy volunteers, not children.
7. Jeukendrup AE, Killer SC. The myths surrounding pre-exercise carbohydrate feeding. Ann Nutr Metab. 2010;57(suppl 2):18-25. doi:10.1159/000322698. doi.org/10.1159/000322698
   Supports the idea that a big carb dose 30–60 minutes before exercise can cause a blood-sugar dip in some people, and that how people feel varies. Review of studies in athletes, not children.
8. Jentjens RLPG, Jeukendrup AE. Prevalence of hypoglycemia following pre-exercise carbohydrate ingestion is not accompanied by higher insulin sensitivity. Int J Sport Nutr Exerc Metab. 2002;12(4):398-413.

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Supports the point that how fast the stomach empties shapes how fast blood sugar rises, and that fat or protein eaten first slows it down. Review of studies in adults.

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Supports the idea that eating vegetables and protein before rice gave a smaller blood-sugar rise. Study in 16 healthy adults.

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Supports the fact that children's brains use an even bigger share: at the peak in childhood, glucose equal to about two-thirds of the body's resting energy.

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Supports the signs of low blood sugar (shaking, sweating, dizziness, confusion, tiredness, irritability) and notes that it is rare in people without diabetes.

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Supports the list of low-blood-sugar symptoms. Written for people with diabetes.

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Supports the signs of low blood sugar in children, and the point that in a healthy child they can be mild, like crankiness.

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Supports the warning signs (much more thirst and peeing, tiredness, hunger with weight loss) and the advice to see the pediatrician right away.

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A 10-year-old's body has about 17 trillion cells.